

Quik SKU Success Prediction Model

Pre-Listing Signal Evaluation | Leave-One-Month-Out Backtest

Jul 2025 · Sep 2025 · Oct 2025 · Nov 2025 · Dec 2025

791

SKUs evaluated
(Availability>70%)

5

Backtest months

30

Pre-listing features

3

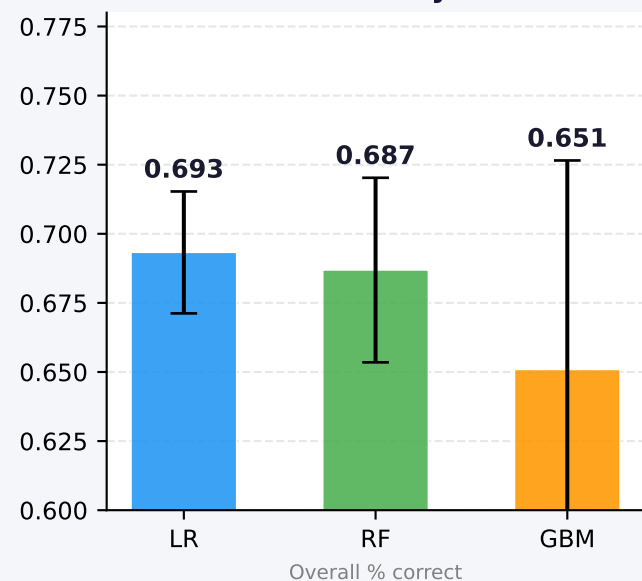
Models benchmarked

★ **Best pre-listing model: Logistic Regression · 69.3% Accuracy · AUC 0.755**

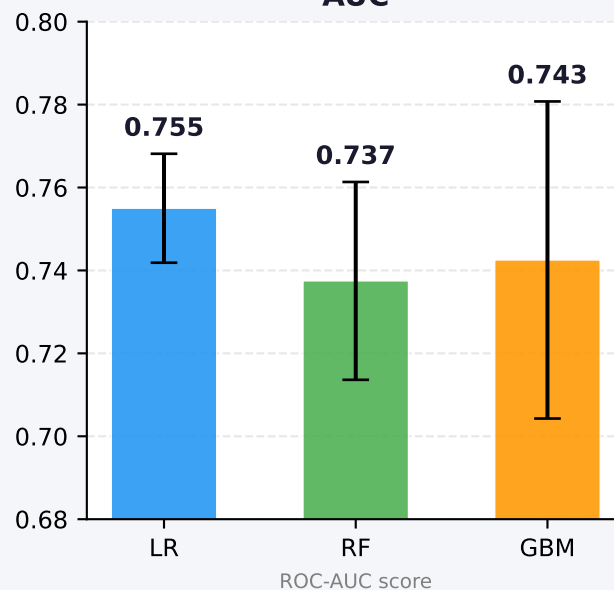
- Objective: Predict whether a newly listed SKU will exceed its ROS target before it goes live on shelf.
- Constraint: No post-listing signals — Availability and observed ROS are deliberately excluded from features.
- Evaluation: Leave-One-Month-Out (LOMO) — train on 4 months, test blind on held-out month.
- Label: $SUCCESS = \max(ROS_m1, ROS_m2) > \max(\min(\text{Bucket P10 Peer ROS}, \text{Vel Floor}), 1)$

Model Performance Comparison (mean $\pm$ std over 5 LOMO folds)

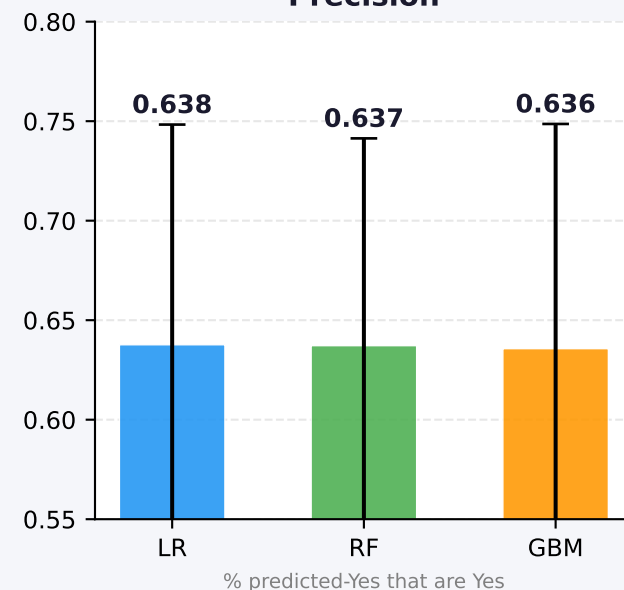
Accuracy



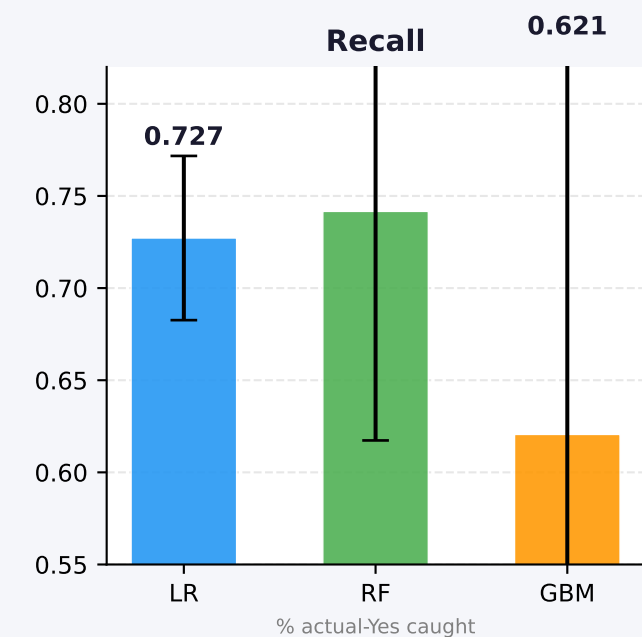
AUC



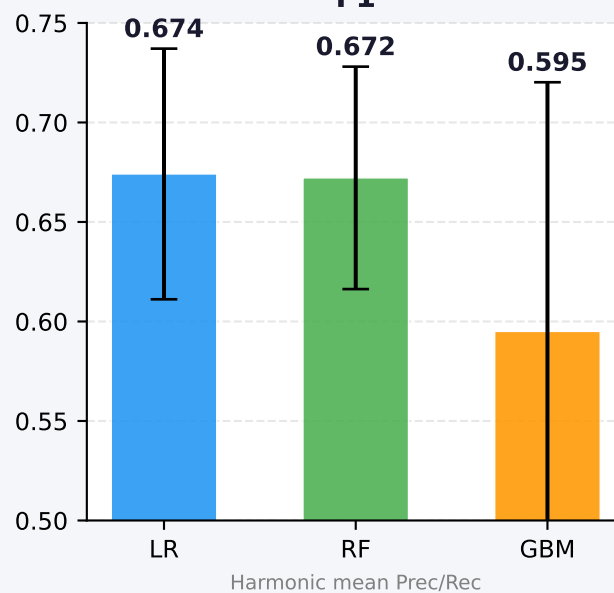
Precision



Recall



F1

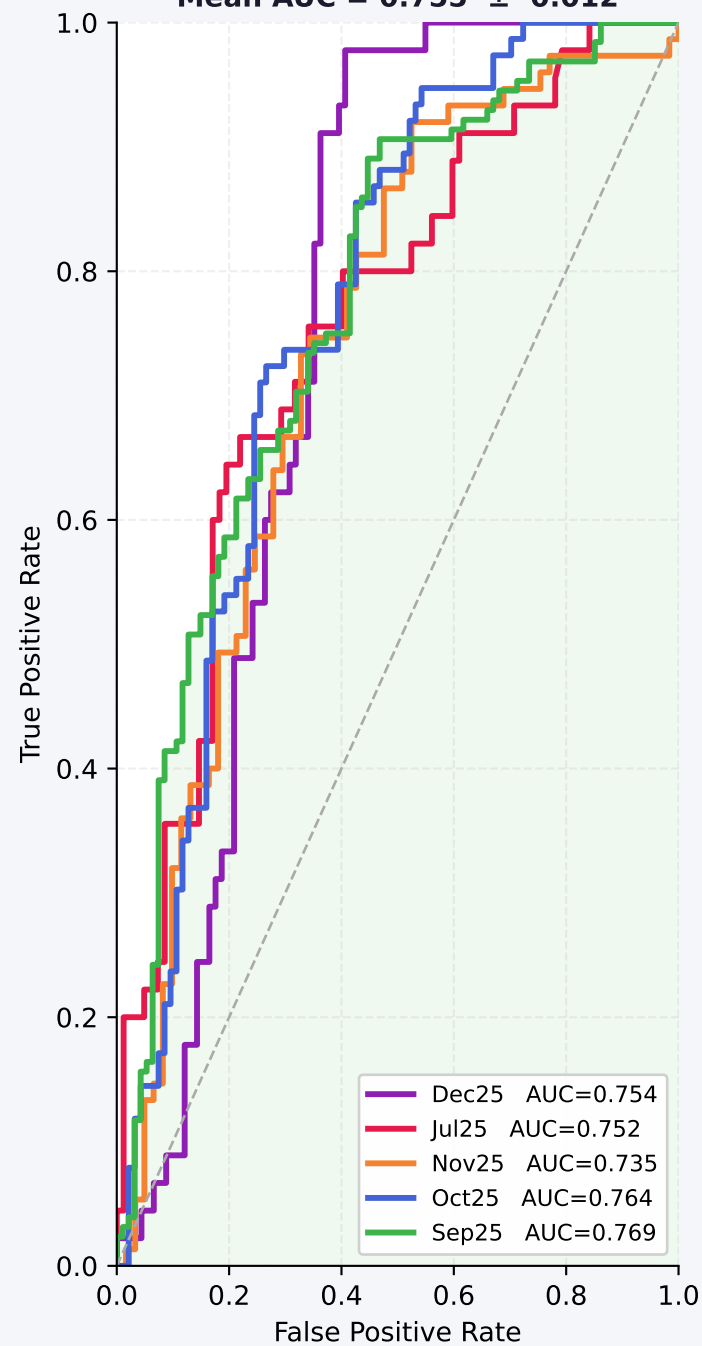


Summary Table

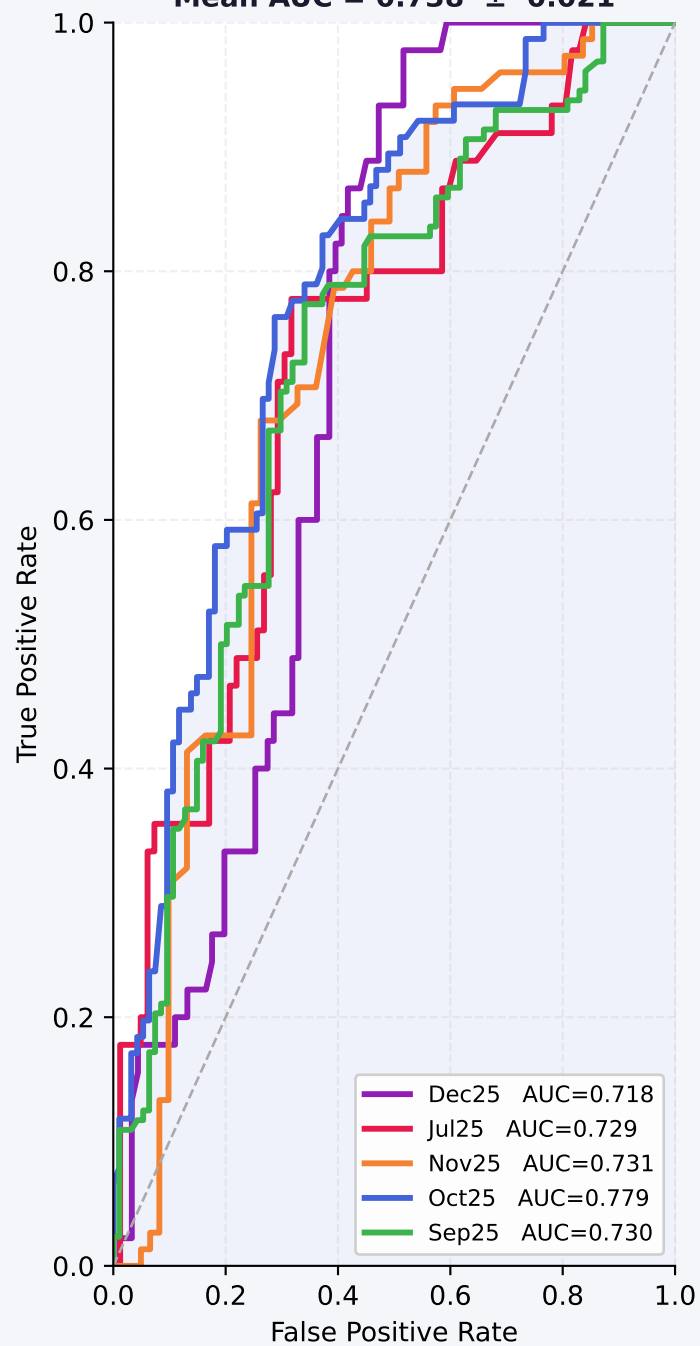
Model	Acc	$\pm$ Std	AUC	F1
LR	0.693	$\pm$ 0.022	0.755	0.674
RF	0.687	$\pm$ 0.033	0.737	0.672
GBM	0.651	$\pm$ 0.076	0.743	0.595

ROC-AUC Curves | Each line = one held-out test month

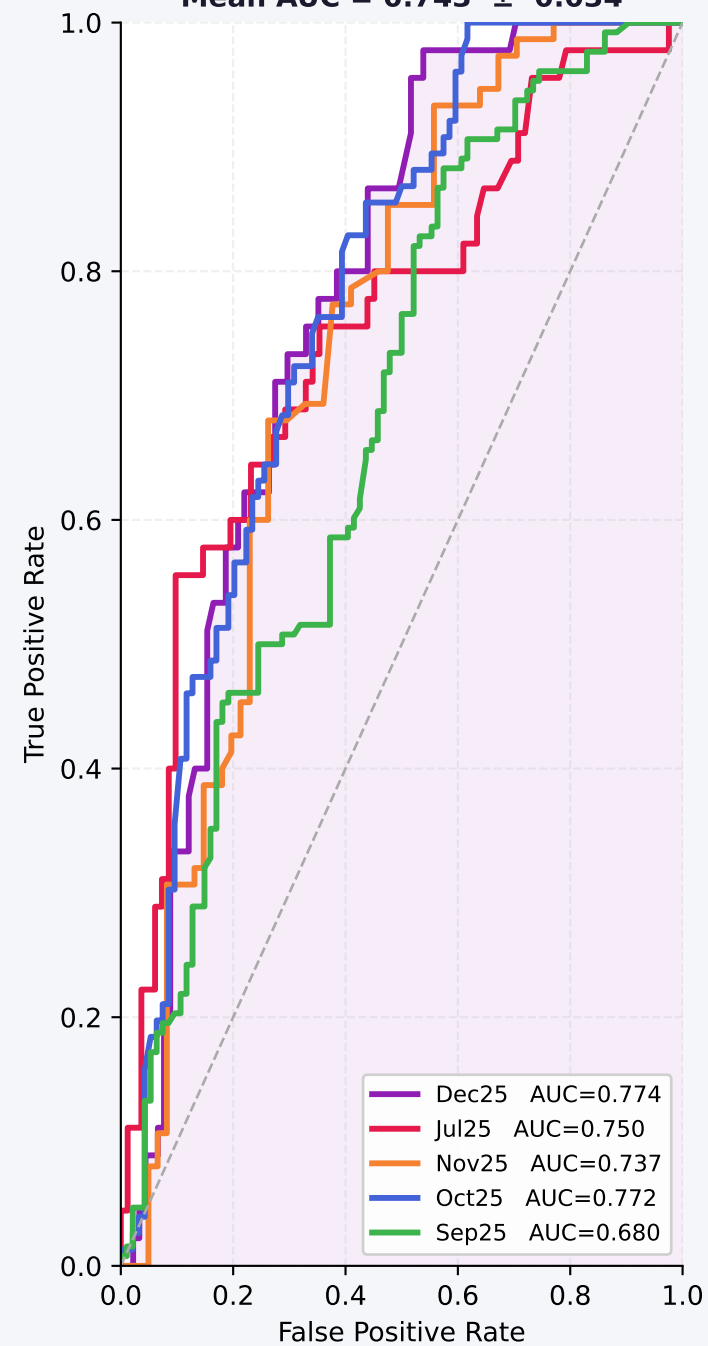
LR — Logistic Regression
Mean AUC = 0.755 ± 0.012



RF — Random Forest
Mean AUC = 0.738 ± 0.021

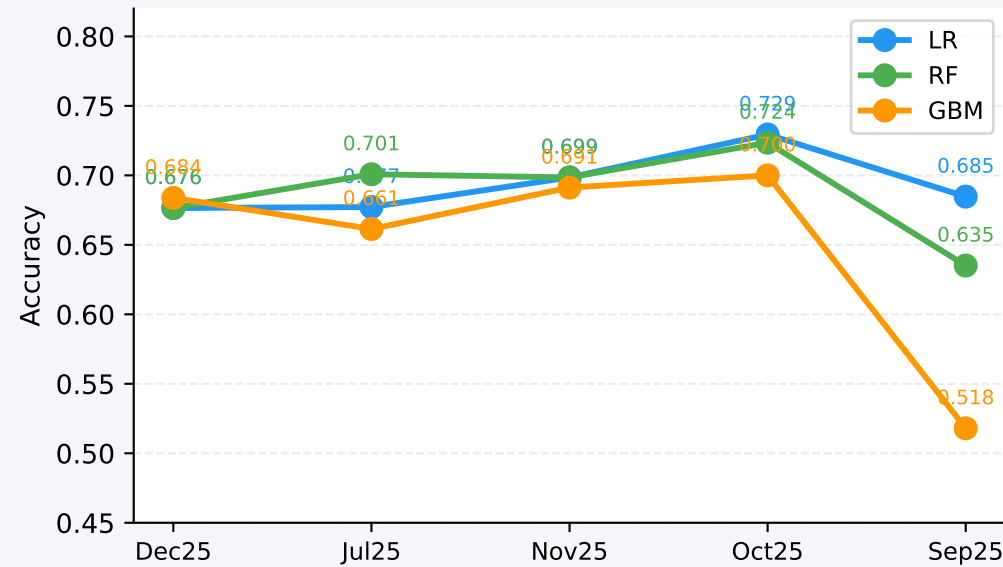


GBM — Gradient Boosting
Mean AUC = 0.743 ± 0.034

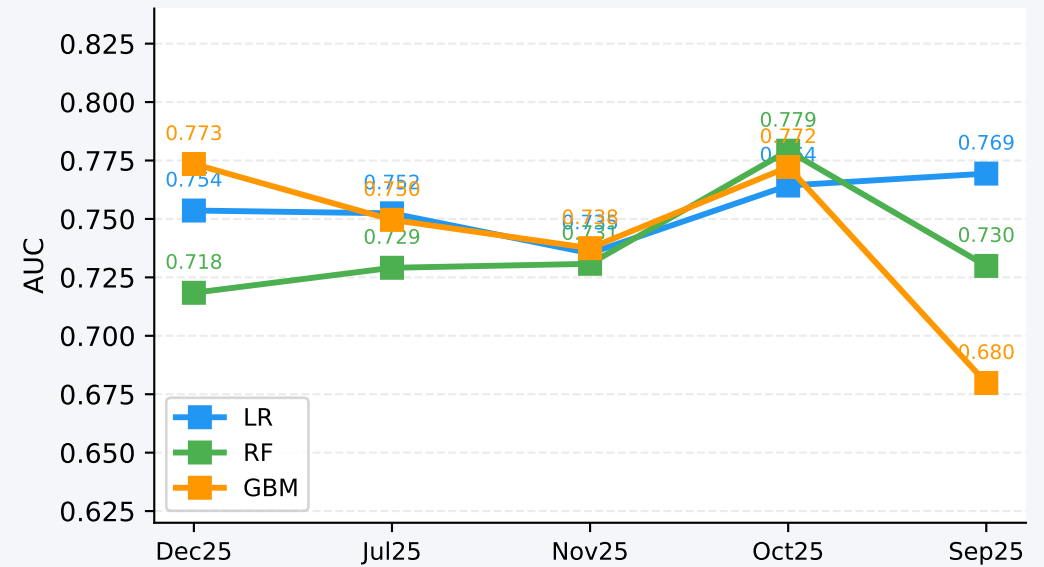


Month-wise Results | LOMO Backtest

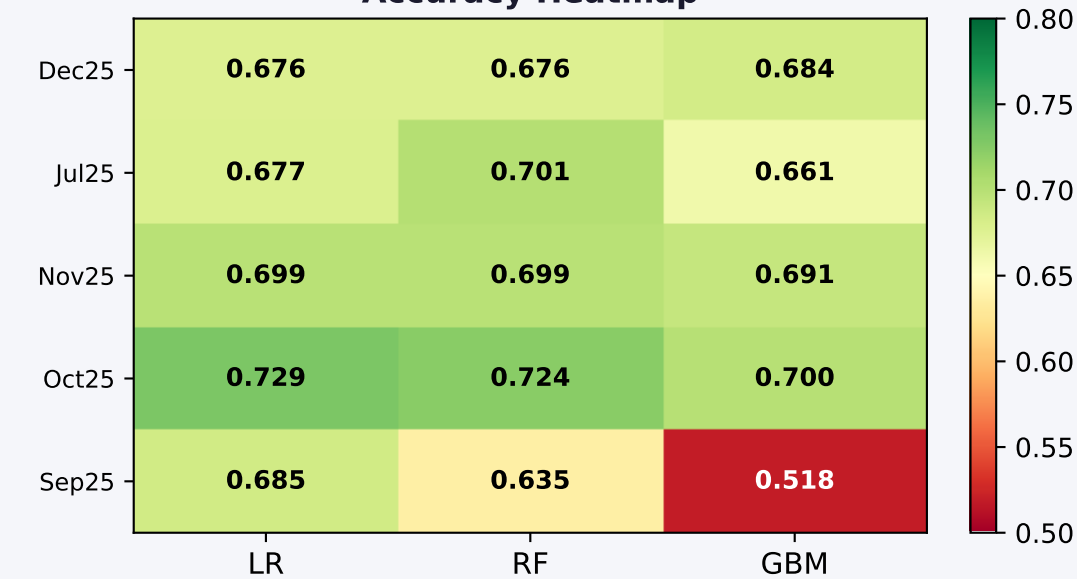
Accuracy per Test Month



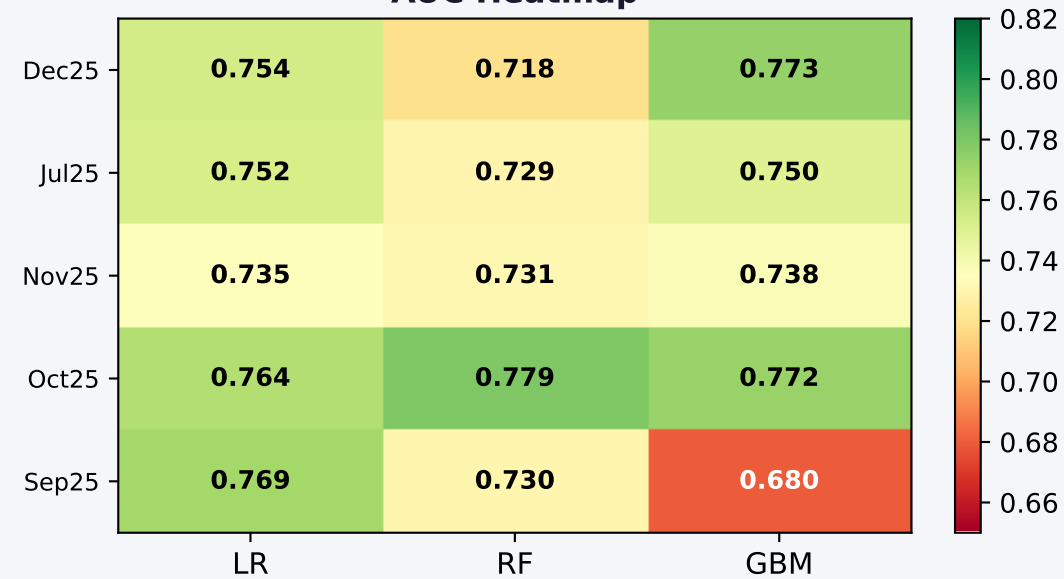
AUC per Test Month



Accuracy Heatmap

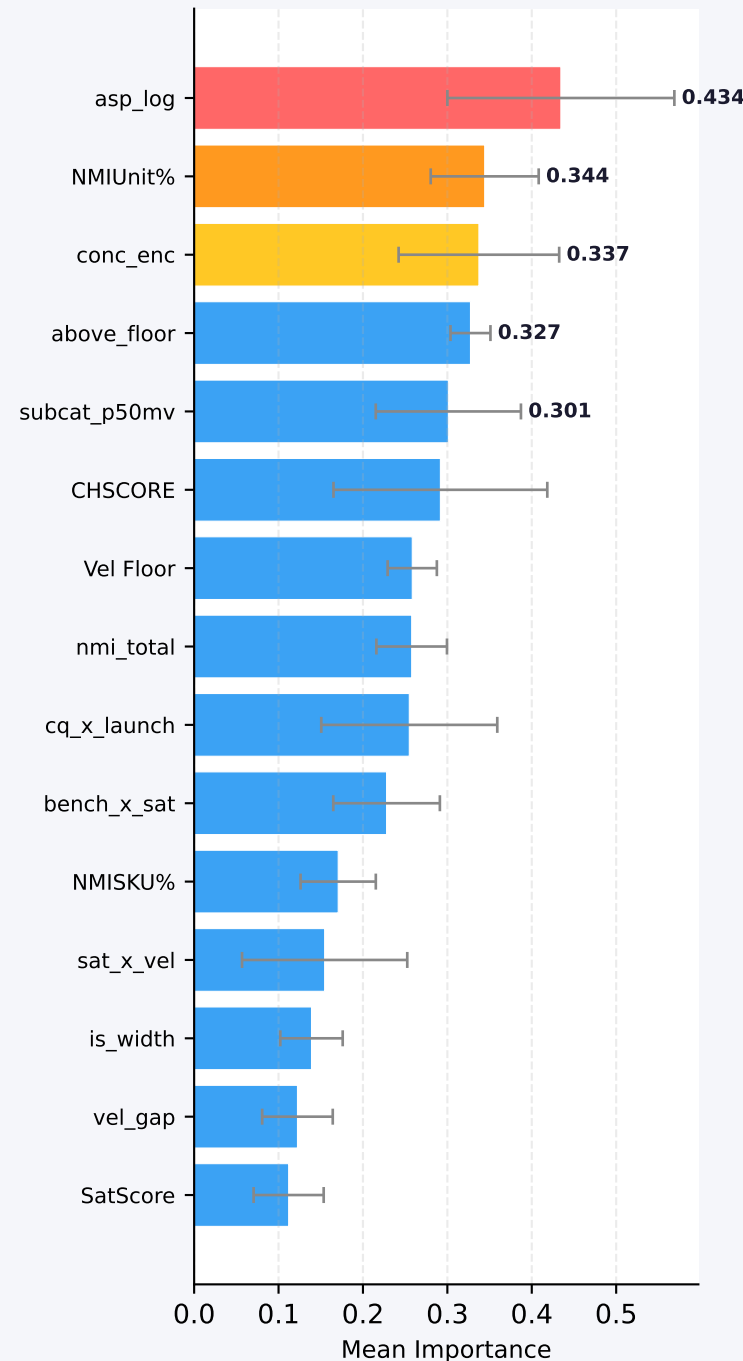


AUC Heatmap

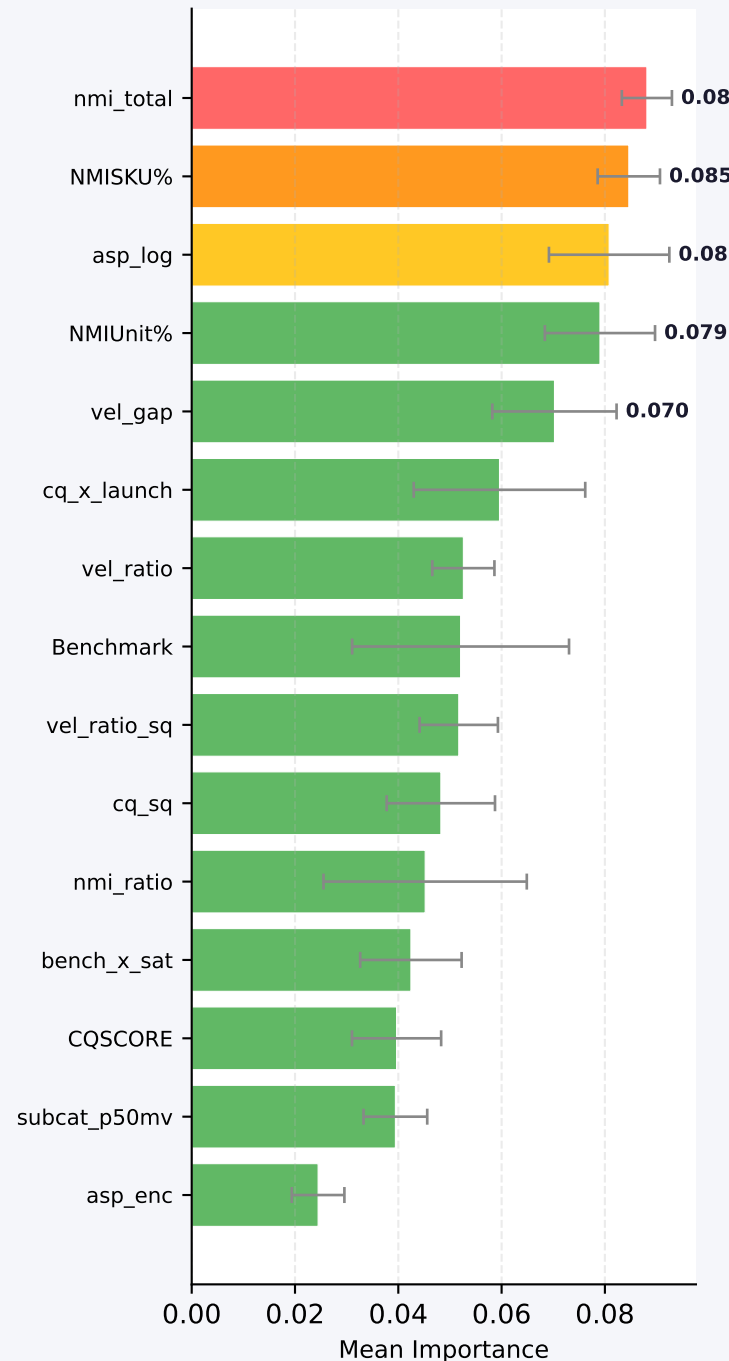


Top Feature Importances | Mean $\pm$ std across all 5 LOMO folds

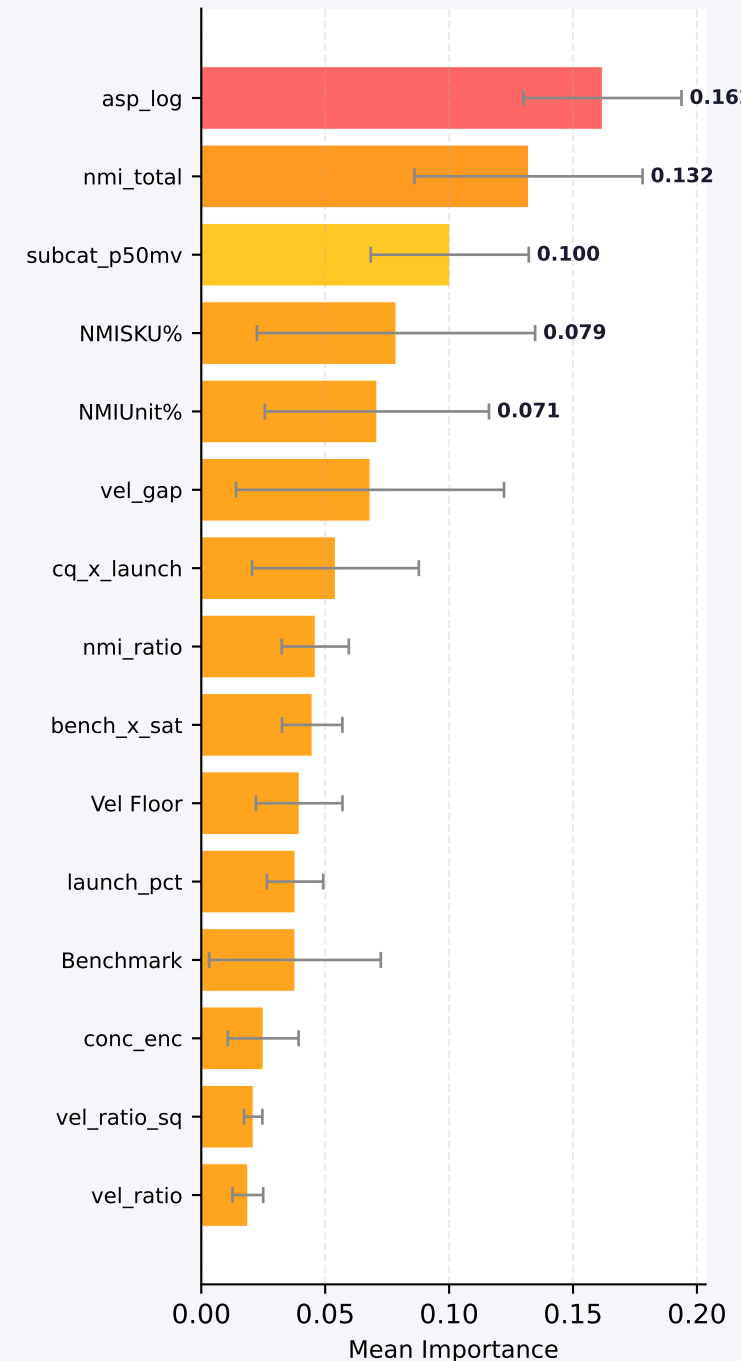
LR — Logistic Regression
(LR: |coef| RF/GBM: MDI importance)



RF — Random Forest
(LR: |coef| RF/GBM: MDI importance)

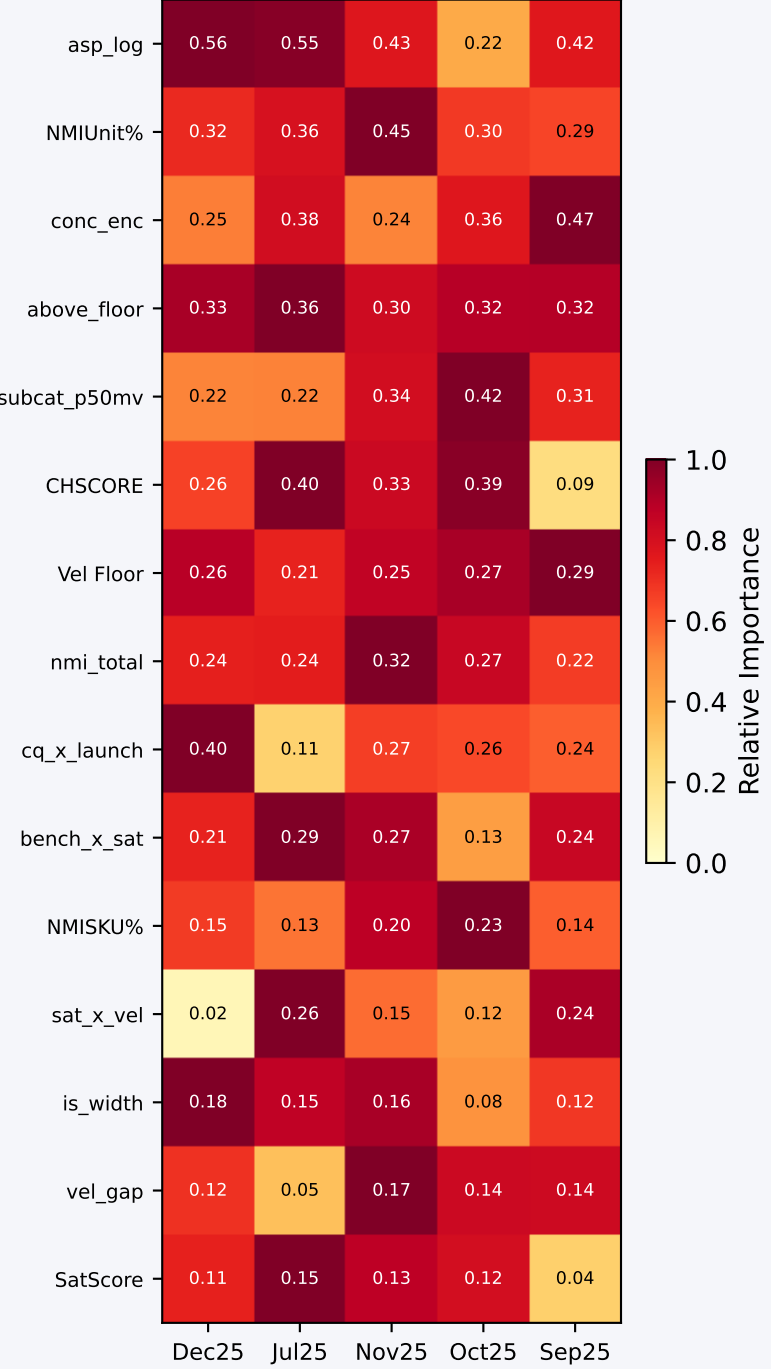


GBM — Gradient Boosting
(LR: |coef| RF/GBM: MDI importance)

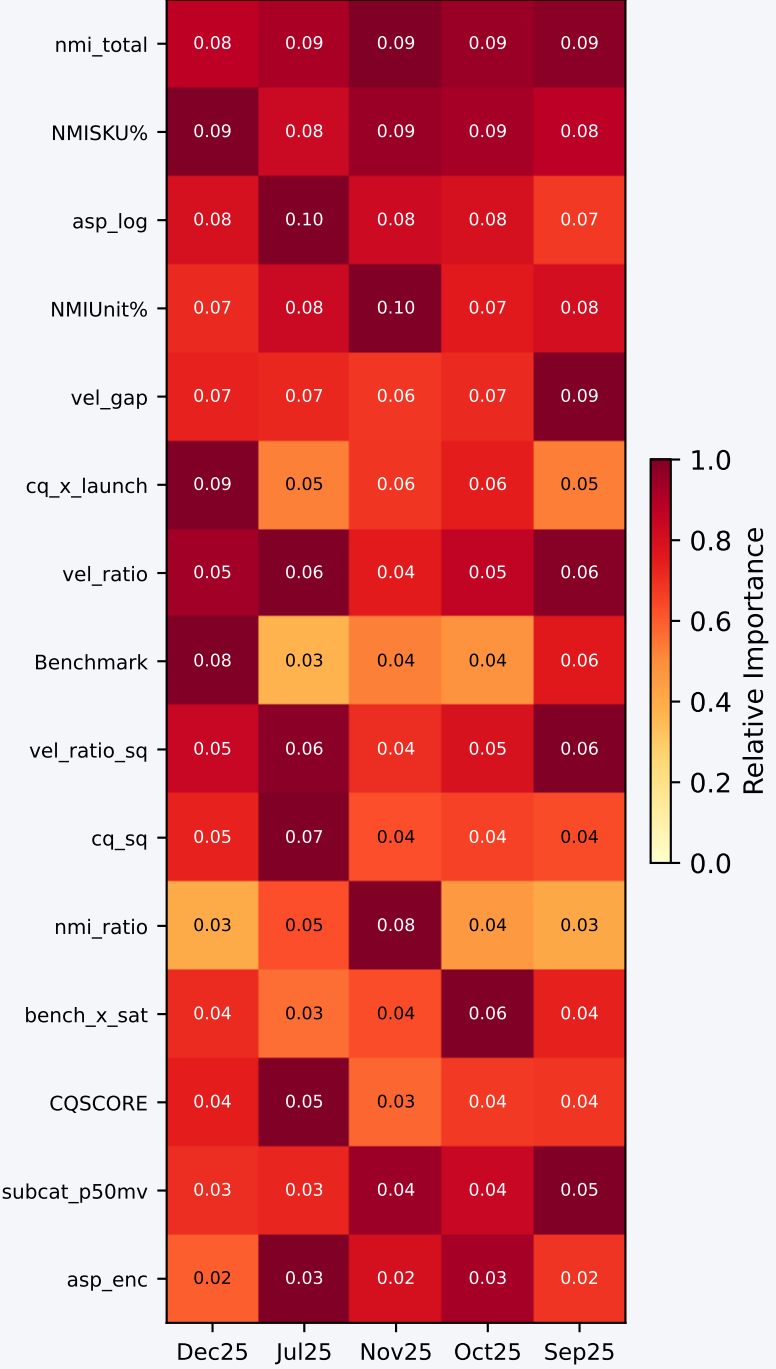


Feature Importance Stability | Top 15 features across all months (row-normalised)

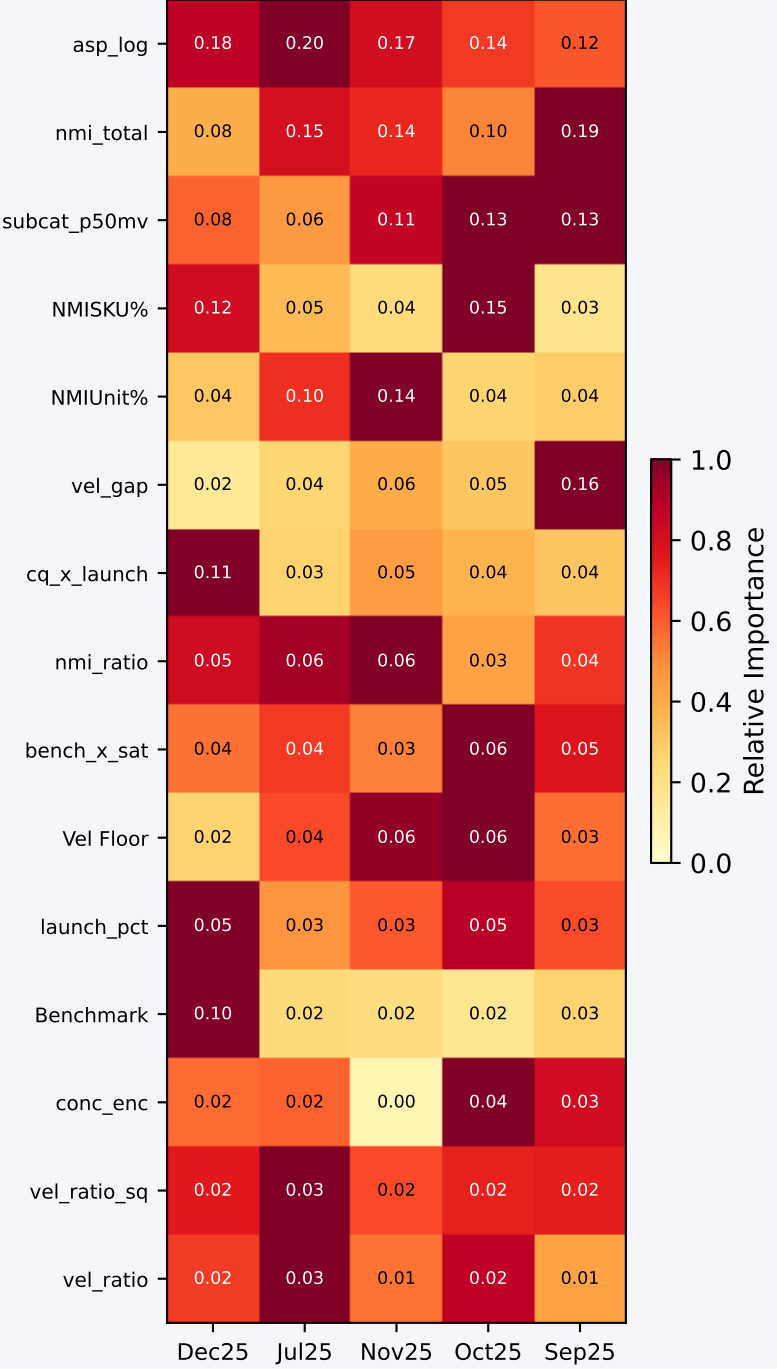
LR — Logistic Regression



RF — Random Forest

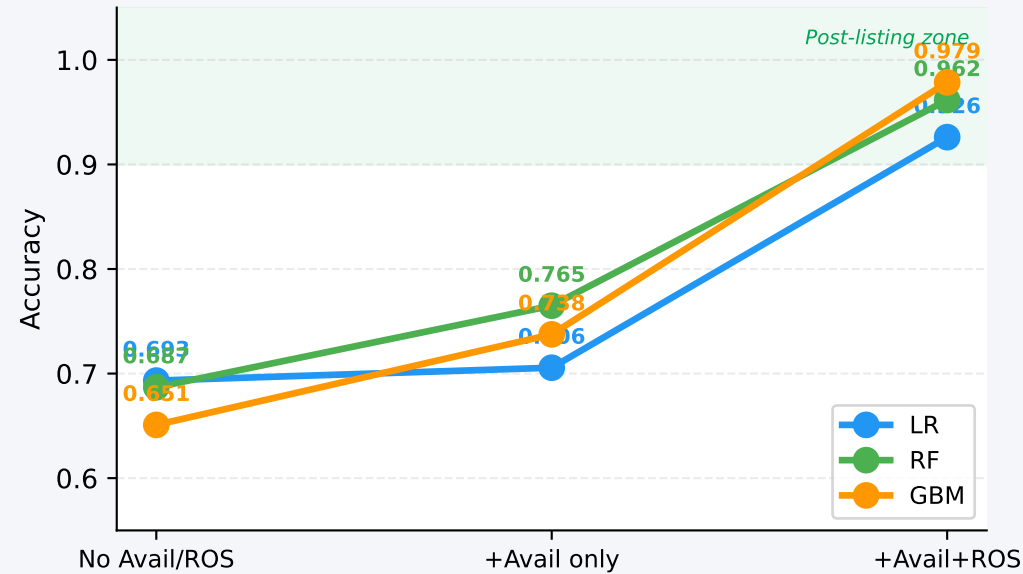


GBM — Gradient Boosting

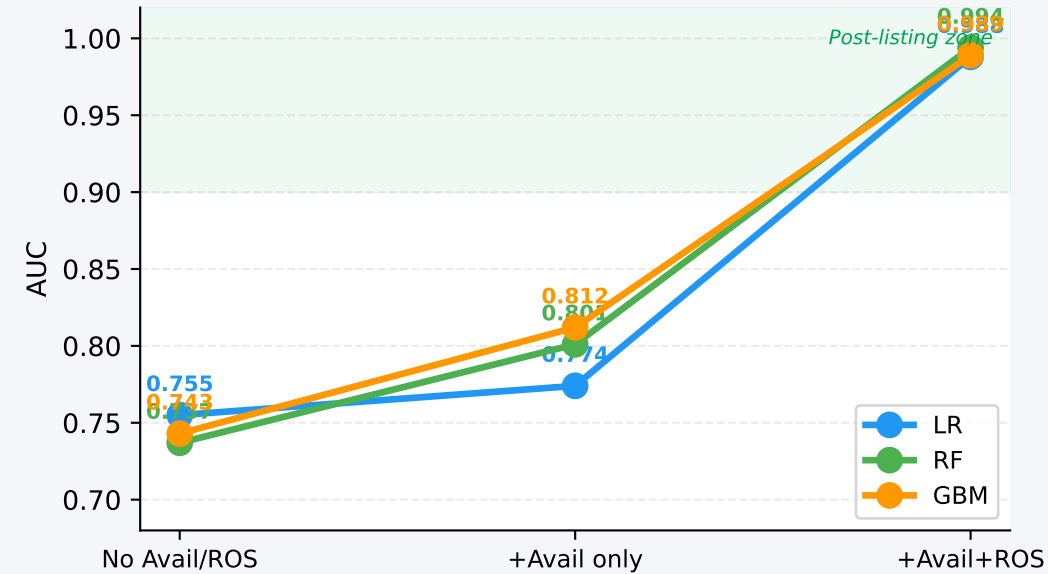


Signal Progression | What each new data source adds

Accuracy — Signal Progression



AUC — Signal Progression



Accuracy Lift by Stage

Model	Pre-listing (No Avail/ROS)	+ Availability (Δ Acc)	+ Avail+ROS (Δ Acc)
LR	0.693	0.706 (+0.012)	0.926 (+0.221)
RF	0.687	0.765 (+0.078)	0.962 (+0.197)
GBM	0.651	0.738 (+0.087)	0.979 (+0.241)

Key Takeaways

- 1. Pre-listing ceiling ~69%**
LR achieves $69.3\% \pm 2.2\%$ using only formula features. This is a data ceiling — without observing the SKU on shelf, prediction accuracy tops out here.
- 2. Availability adds 4-8%**
Tree models capture the non-linear step at the 70% gate. RF: +7.8%, GBM: +8.7%, LR: +1.2%. Trees outperform LR when Avail is included.
- 3. Observed ROS is near-oracle**
 $\max(\text{ros_m1}, \text{ros_m2})$ is the formula numerator. Adding it pushes GBM to 97.9%. Useful for post-listing delisting/ranking decisions.
- 4. LR is the production-safe choice**
GBM collapses on Sep25 (51.8%), std $\pm 7.6\%$. LR is stable $\pm 2.2\%$ and interpretable — best choice for a pre-listing scoring system.